Phenotypes of UK Immunisation Modelling

Balancing Structural Transparency and Policy Engagement in Outbreak and Pandemic Vaccination, 2015-2025

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Introduction

Vaccination campaigns against infectious diseases are among public health's most powerful tools, yet true success hinges on intricate systems of supply chains, cold-chain logistics, workforce mobilisation and community engagement. Delays or inefficiencies in any component can trigger a cascade of unintended consequences; reinforcing feedback loops that widen health disparities and erode trust in health authorities (Rutter et al., 2017). Modern operations research (OR) and modelling techniques (spanning optimisation, simulation and decision-analysis frameworks) offer the rigour to both interrogate and (attempt to) map these complex systems, *ideally* enabling "what-if" exploration of alternative strategies, identification of logistical bottlenecks and probabilistic assessment of campaign outcomes under uncertainty. By integrating epidemiological inputs with resource constraints and behavioural parameters, OR studies have guided vaccine distribution, prioritisation and outreach in diverse settings, ultimately enhancing the resilience and equity of immunisation efforts (Bradley et al., 2017).

Accordingly, it may be understood how within the global landscape, rapid, large-scale vaccination approaches can be intensely specific to ecological and contextual aspects. As such, this scoping review asks: which OR and modelling studies since 2015 have underpinned UK outbreak and pandemic vaccination responses, and to what extent do their methods combine methodological robustness with explicit policy engagement? The UK is considered here both to ensure studies are ecologically comparable, and to narrow the available research to achieve more concise specificity. This report intends to map the spectrum of quantitative approaches, critically assess their assumptions, data sources and validation protocols, and examine pathways of stakeholder involvement and evidence uptake. Framed by a population–concept–context schema; UK immunisation initiatives (population); quantitative models of logistics, coverage and impact with scrutiny of rigour and policy interaction (concept); and UK health-system structures, from national immunisation programmes to emergency response frameworks (context), this review aims to chart how modelling methodologies have informed rapid vaccination campaigns and highlight avenues to reinforce both their realism and policy relevance.

Methodology

A search was conducted in Scopus, which was selected for its broad interdisciplinary coverage (Visser et al, 2021), and relative to the small scale of this scoping review. The search strategy combined controlled vocabulary and free-text terms relating to vaccination or immunisation, modelling or operations-research methods, and outbreak, epidemic, or pandemic contexts (*see Annex A*). All retrieved records (N = 204) were exported for screening. No automated deduplication was required, as no duplicates were identified based on title and authorship.

Studies were included based on the following criteria: (1) the study employed an operations-research or modelling approach to support, evaluate, or propose rapid and large-scale vaccination strategies; (2) the vaccination campaign or immunisation programme modelled targeted human populations and was situated in an outbreak, epidemic, or pandemic context; (3) the authors reported outcomes relevant to operational effectiveness, efficiency, or feasibility; (4) the study explicitly discussed implementation, logistics, policymaking, stakeholder engagement, or other decision-making aspects (e.g., consultation with policymakers, knowledge-translation activities); (5) the research was relative to the United Kingdom (including any constituent countries) and (6) the article was a full-length, peer-reviewed journal research article written in English and published between January 2015 and April 2025. Grey literature, studies within the veterinary subject area, conference abstracts, and dissertations were excluded to maintain quality and consistency. Studies were screened first by title and abstract (N = 204) and subsequently by full-text review (N = 49), with 6 final studies meeting all eligibility criteria being included in the review (*figure 1*).

Model quality was appraised with an amended version of the Philips checklist (Philips et al, 2004), a widely endorsed and actively utilised tool for appraising the quality of decision-analytic models (Oguta et al, 2025; Nguyen et al, 2024). NICE (2024) suggests this specific checklist for more detailed assessments of quality relative to modelling specific studies. Accordingly, items mapping onto three pre-specified domains - (i) structural transparency (six points), (ii) input-data realism (six points), and (iii) validation/output quality (four points) - were retained. Each study was scored relative to each checklist item, where 0 = not met, 1 = partially, 2 = fully. Scores were summed per domain to generate three quantitative indices.

Policymaker engagement was assessed separately, with papers being assigned 'low', 'medium' or 'high' subjective engagement accordingly. Principal-component analysis (PCA) on the three indices plus the engagement domain was performed to include an empirical synthesis of both methodological rigour and policy relevance. Full item wording and scoring sheets are provided in Annex A.

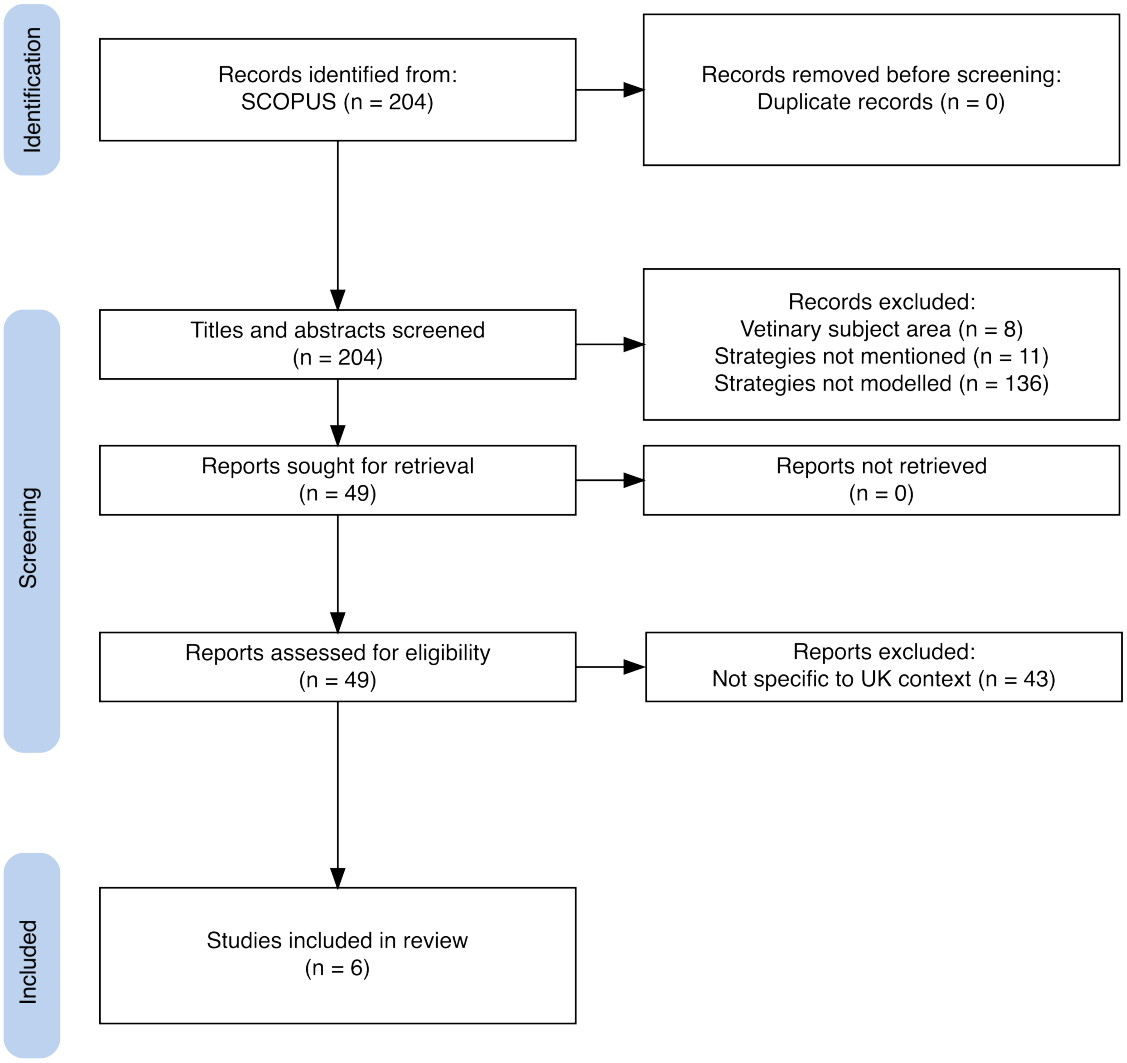

Figure 1: PRISMA Flowchart of Report Selection

Results

Table 1. Study Characteristics

Authors (year)	Vaccine(s) considered	Modelling approach(es)	Population/Setting
Panovska-Griffiths J et al. (2019)	Pandemic influenza vaccines: pre-purchase vs responsive purchase	Dynamic transmission + economic net-present-value analysis under 4 pandemic scenarios	United Kingdom preparedness; varying transmissibility & severity
Panovska-Griffiths J et al. (2020)	Mass immunisation vs antivirals for pandemic influenza	Adapted seasonal-flu model + bespoke health-economic module	Generic UK-style population; wide pandemic parameter space
Pople D et al. (2022)	Pfizer-BioNTech, AstraZeneca, Moderna COVID-19 vaccines in HCWs	Prospective cohort (SIREN) + individual-based transmission model estimating averted infections	Hospital healthcare workers, England second wave
Sonabend R et al. (2021)	COVID-19 vaccines (Pfizer, AstraZeneca, Moderna); Delta variant	Age-structured transmission model with multi-strain dynamics	England; roadmap out of lockdown during Delta wave
Zhang Y et al. (2021)	COVID-19 vaccines (effectiveness under different interventions)	Python-based prognostic IHS dynamic model; visual decision tools	United Kingdom; policy-scenario simulations with multiple NPIs
Zhang S et al. (2023)	COVID-19 vaccines (Pfizer-BioNTech, Oxford-AstraZeneca, Moderna); multiple variants & interventions	Lightweight SEMCVRD model (SEIR extension with vaccinated & severity compartments), mobile-app data integration; privacy-preserving real-time simulation	United Kingdom; calibrated to real case, death & vaccination data; scenario analyses of phased intervention policies

The search yielded six UK focused vaccination modelling studies published between 2015 and 2025 (*Table 1*). Two papers by Panovska Griffiths et al. address influenza preparedness under varied transmissibility and severity: the 2019 study pits pre purchase against responsive procurement strategies via a dynamic SEIR-economic model and net present value analysis across four pandemic scenarios, while the 2020 work adapts a seasonal flu transmission framework with a bespoke health economic module to compare mass immunisation with antiviral deployment over a wide parameter space in a generic UK style setting. The remaining four studies centre on COVID 19. Pople et al. (2022) merge SIREN prospective cohort data from hospital healthcare workers in England's second wave with an individual based transmis-

sion model to estimate averted infections for Pfizer BioNTech, AstraZeneca, and Moderna vaccines. Sonabend et al. (2021) deploy an age structured, multi strain model-calibrated for Delta dynamics-to evaluate England's lockdown exit roadmap under alternative vaccine roll out scenarios. Zhang et al. (2021) introduce a Python based prognostic IHS dynamic model equipped with interactive visual decision tools for simulating vaccine effectiveness amid various non pharmaceutical interventions, and Zhang et al. (2023) present a lightweight SEMCVRD extension of SEIR-adding vaccinated and severity compartments and integrating mobile app data for privacy preserving, real time simulation-calibrated to observed case, death, and vaccination data for phased policy analyses.

Table 2. Amended Philips et al. Quality Appraisal

Authors (year)	**Σ Structure score (%)**	**Σ Data score (%)**	**Σ Consistency score (%)**	**Policy score**
Panovska-Griffiths et al. (2019)	0.83	0.33	0.25	2
Panovska-Griffiths et al. (2020)	1.00	0.33	0.25	3
Pople et al. (2022)	1.00	0.50	0.50	2
Sonabend et al. (2021)	1.00	0.66	0.75	3
Zhang et al. (2021)	1.00	0.16	0.00	1
Zhang et al. (2023)	1.00	0.50	0.75	1

*Full table of individual scores that comprise each category (structure, etc) in Annex A

*Policy score given directly, with 1 indicating low and 3 indicating high engagement

Applying the amended Philips et al. checklist (*Annex A*) generated three continuous indices-Structure, Data realism and Consistency-and a discrete Policy engagement score. Structure was uniformly high (1.00) except for Panovska Griffiths 2019 (0.83), which, despite clearly framing its decision problem and tabulating SEIR-economic equations, offered only brief justification for homogeneous mixing and fixed efficacy assumptions. All other studies fully described their model equations and simplifications: Panovska Griffiths 2020 justified each differential equation assumption; Pople et al. tabulated transmission routes and vaccine efficacy parameters alongside cohort design; Sonabend et al. detailed multi strain, waning immunity extensions with full priors; Zhang 2021 specified automata parameters; and Zhang

2023 documented every SEMCVRD compartment and app integration. Data realism scores ranged from 0.16 for both Panovska Griffiths papers (each quoting median R_0 and CFR without a formal search strategy and assuming point estimate efficacies) to 0.66 for the other five studies, which, despite sourcing parameters from cohort, registry, surveillance or literature and anonymised EMRS uploads, similarly lacked explicit search protocols. Consistency varied more widely: the Panovska Griffiths models performed only deterministic scenario sweeps with no internal checks; Pople et al. reported interquartile ranges from stochastic runs but no joint parameter PSA; Zhang 2021 omitted PSA and validation; Zhang 2023 conducted point estimate error checks and quantitative comparisons; Sonabend et al. however included full Bayesian inference, credible intervals, goodness of fit plots and qualitative benchmarking (0.75). Policy engagement ratings-based on narrative accounts of stakeholder collaboration-span from high (Panovska Griffiths' close PHE/DHSC partnerships and Sonabend's direct roadmap input) through moderate (Pople's use of national surveillance with limited consultation) to low (the Zhang papers' positioning as decision tools without formal liaison).

An initial unrotated extraction identified two eigenvalues >1 (2.20 and 1.03), incurring the retention of two factors, and the subsequent varimax rotation yielded loadings as shown in Table 3. The first dimension explained 54.9% of the variance and was dominated by ΣData (0.978) and ΣConsistency (0.914), with a secondary loading from Policy engagement (0.545), showcasing a potential methodological rigour axis. The second dimension accounted for 25.7% of the variance, loading positively on ΣStructure (0.775) and negatively on Policy (-0.622); this might be interpreted as a potential structural transparency versus policy orientation dimension.

Table 3. Varimax-rotated PCA loadings, variance explained and component scores

Item / Study	Dimension 1	Dimension 2
Rotated loadings		
Σ Structure	-	0.775
Σ Data	0.978	-
Σ Consistency	0.914	-
Policy engagement	0.545	-0.622
SS loadings	2.195	1.028
Proportion Var (%)	54.9	25.7
Cumulative Var (%)	54.9	80.6
Component scores by study		
Panovska-Griffiths et al.(2019)	-0.661	-1.581
Panovska-Griffiths et al.(2020)	-0.079	-0.476
Pople et al.(2022)	0.378	0.328
Sonabend et al.(2021)	1.438	-0.288
Zhang et al.(2021)	-1.478	0.853
Zhang et al.(2023)	0.402	1.164

*Rotated loadings: Only values ≥ 0.40 are shown.

*SS loadings (sum of squared loadings), Proportion Var and Cumulative Var summarise the variance captured by each component.

*Component scores by study locate each paper in the dimension 1-2 space (see Figure 2).

Component scores (Table 3) position each study within the two dimensional appraisal space. Sonabend 2021 scores highest on the first dimension (+1.44) and moderately negative on the second (-0.29), indicating rigorous data realism and validation with limited structural focus. Pople 2022 and Zhang 2023 occupy positive space on dimension 1 (0.38 and 0.40) but diverge on the second (+0.33 vs +1.16), reflecting shared data strength tempered by varying structural transparency. The two Panovska Griffiths studies cluster at negative levels of the first dimension (-0.66 and -0.08) and at strongly negative values of the second (-1.58 and -0.48), signalling modest empirical underpinning yet high policy engagement. Zhang 2021 lies at the lowest methodological extreme (dimension 1 = -1.48) with a positive structural emphasis (dimension 2 = 0.85).

K means clustering (k = 2) was performed on each study's rotated component scores to partition the six papers into two empirically driven profiles (Figure 2). Here, each study is represented by its coordinates on the first principal component (methodological rigor) and second principal component (structural transparency versus policy orientation), and

grouped to minimise within cluster variance. The lower cluster comprises Panovska Griffiths (2019), Panovska Griffiths (2020) and Sonabend (2021); these three studies share negative dimension 2 values, reflecting comparatively stronger documented stakeholder engagement (Policy $\geq$ 0.40 2) than exhaustive model structure exposition, and yet span a wide range on dimension 1 (from -0.66 to +1.44), with Sonabend (2021) standing out for its combination of deep policy alignment and high data realism/validation. By contrast, the upper cluster contains Zhang (2021), Zhang (2023) and Pople (2022), all of which achieve greater structural transparency (Σ Structure = 1.00) but exhibit variable values on dimension 1 (1.48 to +0.40), indicating that full architectural clarity does not necessarily coincide with stronger empirical underpinning.

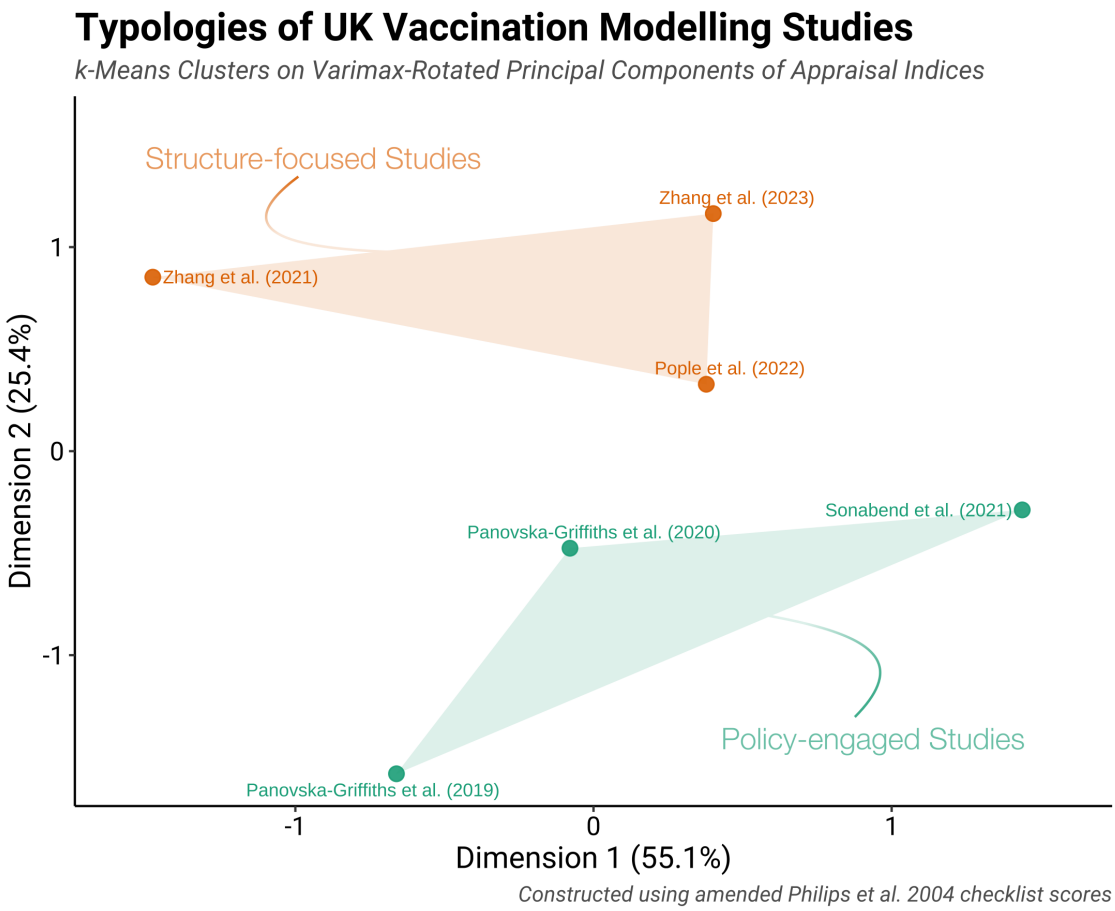

Figure 2: Study Phenotypes Based on PCA and k-Means Clustering

To make these groupings visually explicit, each cluster is enclosed by a convex hull polygon drawn around its most extreme dimension 1 and 2 points, clearly demarcating the two methodological engagement 'phenotypes'. Although there is within cluster heterogeneity in dimension 1, the dominant axis of separation lies along dimension 2: the "policy engaged" cluster occupies the lower half of the plot, while the "structure focused" cluster lies in the upper half. This top versus bottom divergence underscores a potential principal trade off in recent UK vaccination modelling studies between prioritising stakeholder consultation and emphasising transparent model architecture, with methodological rigor providing nuance within each profile.

Discussion

This review mapped how operations research and modelling studies underpinned UK rapid vaccination responses. A 'policy engaged' cluster was identified, excelling in stakeholder collaboration and relevance but relying on deterministic sweeps and offering partial model transparency, and a 'structure focused' cluster was identified, with transparent architectures and mechanistic detail yet weaker policymaker liaison. Principal component analysis indicated that data realism and validation aligned with methodological rigour, while structural transparency versus policy orientation marked a primary divergence in the literature.

Although many models facilitated "what if" exploration (identifying bottlenecks and projecting outcomes under uncertainty) their heterogeneity complicated synthesis. Deterministic SEIR sweeps rapidly scan parameter spaces but may under represent uncertainty, whereas Bayesian inference and probabilistic sensitivity analyses require more data and computing resources but might better quantify risk. Accordingly, modelling of cold chain constraints, workforce mobilisation and community uptake would benefit from hybrid frameworks marrying stochastic calibration with open source transparency.

Several limitations temper conclusions; restricting the search to Scopus and UK outbreak contexts may have overlooked insights from other databases, geographies or routine immunisation programmes where supply chain challenges differ. Further, the small final sample

of six heterogeneous studies may have limited comparability. Finally, this review does not consider that high policy engagement scores often reflect long-standing institutional ties to public health bodies (PHE, DHSC), potentially serving to provide a barrier for less connected researchers that risks entrenching inequities in model access and influence.

To address the structural transparency versus policy orientation trade off, this review recommends four actions. First, journals and funders should require a "Stakeholder Engagement" section in modelling reports, detailing consultation processes and decision maker roles. Second, research teams should convene co design workshops with all stakeholders to co define scope and assumptions. Third, rapid response models should include full probabilistic sensitivity analysis and real time data assimilation to strengthen uncertainty quantification. Finally, developing open source platforms with templates and stakeholder liaison toolkits would democratise access to rigorous modelling and improve transparency. Ultimately, UK immunisation models should apply rigour equally to logistical simulation and policy engagement.

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Annex A

Amended Philips et al. Checklist

Code	Checklist item	Abbreviated checklist item	Score range	Σ
S1	There should be a clear statement of the decision problem prompting the analysis.	Decision problem and objective clearly stated	0–2	ΣS
S4	All structural assumptions should be transparent and justified.	Structural assumptions transparent and justified	0–2	ΣS
S6	The appropriate model type will be dictated by the stated decision problem and the choices made regarding the causal relationships within the model.	Model type appropriate to the question	0–2	ΣS
D1	Methods for identifying data should be transparent and it should be clear that the data identified are appropriate given the objectives of the model.	Systematic, documented data identification	0–2	ΣD
D2b	The methods and assumptions that are used to extrapolate short-term results to final outcomes should be documented and justified.	Treatment effect synthesis / extrapolation justified	0–2	ΣD
D4d	Probabilistic analysis is the most appropriate method of handling parameter uncertainty because it facilitates assessment of the joint effect of uncertainty over all parameters.	Parameter uncertainty handled (ranges or PSA)	0–2	ΣD
C1	There should be evidence that the internal consistency of the model has been evaluated in terms of its mathematical logic.	Internal logic / extreme value checks performed	0–2	ΣC
C2	The results of a model should be compared with those of previous models and any differences should be explained.	External calibration or comparison with other models	0–2	ΣC
P1	NA	Level of policymaker engagement	1-3	ΣP

*0 = no/unclear, 1 = partial, 2 = fully met

*Checklist item wording is directly taken from Phillips et al. checklist for all items other than P1, which is added for the purposes of this review.

*P1 has each paper ranked according to their corresponding level of policymaker engagement, with 1 indicating low engagement, and 3 indicating high engagement.

Full Checklist Scores

Authors (year)	**S1**	**S4**	**S6**	**D1**	**D2b**	**D4d**	**C1**	**C2**	**P1**
Panovska-Griffiths et al. (2019)	2	1	2	1	0	1	0	1	2
Panovska-Griffiths et al. (2020)	2	2	2	1	0	1	0	1	3
Pople et al. (2022)	2	2	2	1	1	1	1	1	2
Sonabend et al. (2021)	2	2	2	1	1	2	2	1	3
Zhang et al. (2021)	2	2	2	1	0	0	0	0	1
Zhang et al. (2023)	2	2	2	1	1	1	1	2	1

Scopus Search Boolean String

```
TITLE-ABS-KEY ( "operations research" OR "operational research" OR
"mathematical model*" OR "simulation" OR "agent-based" OR "statistical model" )
AND TITLE-ABS-KEY ( "vaccination campaign*" OR "vaccine rollout*" OR
"mass vaccin*" OR " program*" OR "vaccination program*" OR
"vaccine distribution" OR "vaccination strateg*" OR "vaccination coverage" ) AND
TITLE-ABS-KEY ( pandemic OR epidemic OR outbreak ) AND TITLE-ABS-KEY ( policy OR
policymaker* OR "decision-maker*" OR "decision making" OR "health system*" OR
logistic* OR "resource allocation" OR implementation OR deployment OR planning
OR feasibility OR "stakeholder engagement" OR "decision conference" OR "co-design"
OR "knowledge translation" OR "evidence-to-policy" ) AND PUBYEAR > 2014 AND
PUBYEAR < 2026 AND ( LIMIT-TO ( SRCTYPE , "j" ) ) AND ( LIMIT-TO ( AFFILCOUNTRY ,
"united kingdom" ) ) AND ( LIMIT-TO ( DOCTYPE , "ar" ) ) AND ( EXCLUDE ( SUBJAREA ,
"vete" ) ) AND ( LIMIT-TO ( LANGUAGE , "english" ) )
```

Principal Component Analysis R Code

```r
# Set up ----
library(psych)
library(dplyr)
library(tidyr)

get_script_path <- function() {
  # RStudio
  if (requireNamespace("rstudioapi", quietly = TRUE)) {
    sp <- rstudioapi::getActiveDocumentContext()$path
    if (!is.null(sp) && sp != "") {
      return(normalizePath(dirname(sp)))
    }
  }
  # Rscript
  args <- commandArgs(trailingOnly = FALSE)
  file_arg <- grep("--file=", args)
  if (length(file_arg) > 0) {
    sp <- gsub("--file=", "", args[file_arg])
    return(normalizePath(dirname(sp)))
  }
  stop("Could not determine the script path. Please consider setting it
  ↪  manually!")
}

script_path <- get_script_path()

# Getting my functions ----
```

```r
setwd(script_path)
source(file = file.path("..", "..", "Old R Stuff", "Function_Library.R"))

ClusterPlot <- function(pca_object, pc1 = 1, pc2 = 2,
                        k = 2,
                        hull = TRUE,
                        title = NULL, subtitle = NULL, caption = NULL,
                        x_label = NULL, y_label = NULL,
                        point_size = 3, alpha_pts = 0.9,
                        equal_axes = FALSE) {
  library(ggplot2)
  library(showtext)
  library(ggrepel)
  showtext_auto()
  font_add_google("Roboto")

  # 1. scores on selected PCs
  ↪  ------------------------------------------------
  scores <- as.data.frame(pca_object$scores[, c(pc1, pc2)])
  colnames(scores) <- c("PC1", "PC2")
  scores$Study <- rownames(scores)

  # 2. k-means clustering
  ↪  ---------------------------------------------------
  set.seed(123)
  cl          <- kmeans(scores[, c("PC1", "PC2")], centers = k)
  scores$cl   <- as.factor(cl$cluster)

  # 3. base plot
  ↪  ------------------------------------------------------------
```

```r
plt <- ggplot(scores, aes(PC1, PC2, colour = cl, fill = cl)) +
  geom_point(size = point_size, alpha = alpha_pts) +
  geom_text_repel(aes(label = Study), size = 3, show.legend = FALSE) +
  scale_colour_brewer(palette = "Dark2") +
  scale_fill_brewer(palette   = "Dark2") +   # MATCH hull/ellipse fill
  guides(fill = "none") +                    # one legend (colour only)
  theme_classic() +
  theme(
    text          = element_text(family = "Roboto"),
    plot.title    = element_text(size = 18, face = "bold"),
    plot.subtitle = element_text(size = 12, face = "italic", colour =
    ↪  "grey30"),
    plot.caption  = element_text(size = 10, face = "italic", colour =
    ↪  "grey30"),
    axis.title    = element_text(size = 14),
    axis.text     = element_text(size = 12, colour = "black"),
    legend.title  = element_blank()
  ) +
  labs(
    title    = title,
    subtitle = subtitle,
    caption  = caption,
    x = ifelse(is.null(x_label),
               paste0("PC", pc1, " (", round(pca_object$values[pc1] /
               ↪  sum(pca_object$values) * 100, 1), "%)"),
               x_label),
    y = ifelse(is.null(y_label),
               paste0("PC", pc2, " (", round(pca_object$values[pc2] /
               ↪  sum(pca_object$values) * 100, 1), "%)"),
```

```r
                y_label)
    )

  # 4. cluster outlines
  ↪  ----------------------------------------------------
  if (hull) {
    hull_df <- scores %>%
      dplyr::group_by(cl) %>%
      dplyr::slice(chull(PC1, PC2))

    plt <- plt +
      geom_polygon(data = hull_df, aes(group = cl),
                   alpha = 0.15, colour = NA)
  } else {
    plt <- plt +
      stat_ellipse(aes(group = cl), geom = "polygon",
                   alpha = 0.15, colour = NA, level = 0.95)
  }

  # 5. equal-axes option
  ↪  ----------------------------------------------------
  if (equal_axes) {
    lim <- max(abs(c(scores$PC1, scores$PC2)))
    plt <- plt + xlim(-lim, lim) + ylim(-lim, lim)
  }
  return(plt)
}

# Amended philips table creation ----
```

```r
txt <- "
Authors|Sigma_Structure|Sigma_Data|Sigma_Consistency|Policy
Panovska-Griffiths et al. (2019)|0.83|0.33|0.25|2
Panovska-Griffiths et al. (2020)|1.00|0.33|0.25|3
Pople et al. (2022)|1.00|0.50|0.50|2
Sonabend et al. (2021)|1.00|0.66|0.75|3
Zhang et al. (2021)|1.00|0.16|0.00|1
Zhang et al. (2023)|1.00|0.50|0.75|1
"

df <- read.table(text = txt,
                 header = TRUE,
                 sep = "|",
                 stringsAsFactors = FALSE,
                 strip.white = TRUE)
rownames(df) <- df$Authors
df$Authors <- NULL

# Data standardisation ----
philips_scaled <- scale(df)

# Unrotated PCA to inspect eigenvalues ----
pca_full <- principal(philips_scaled, nfactors = ncol(df), rotate = "none")
eigs <- pca_full$values

# number of components (eigenvalue > 1) ----
ncomp <- sum(eigs > 1)

# scree plot ----
```

```r
fa.parallel(philips_scaled, fa = "pc", n.iter = 100)

# Varimax-rotated PCA with chosen components ----
philips_pca <- principal(philips_scaled, nfactors = ncomp, rotate =
↪  "varimax")

# Rotated loadings and variance ----
print(philips_pca$loadings, cutoff = 0.4)
eigenvals <- philips_pca$values
variance <- eigenvals / sum(eigenvals)
print(round(variance * 100, 1))  # % variance per component

# Component scores per study ----
pca_scores_var <- as.data.frame(philips_pca$scores)
rownames(pca_scores_var) <-rownames(df)
print(pca_scores_var)

# Biplot of rotated PCA ----
cluster_plot <- ClusterPlot(
  pca_object = philips_pca,
  k          = 2,
  hull       = TRUE,
  x_label = "Dimension 1 (55.1%)",
  y_label = "Dimension 2 (25.4%)",
  title      = "Typologies of UK Vaccination Modelling Studies",
  subtitle   = "k-Means Clusters on Varimax-Rotated Principal Components of
  ↪  Appraisal Indices",
  caption    = "Constructed using amended Philips et al. 2004 checklist
  ↪  scores",
```

```r
  equal_axes = TRUE
)

print(cluster_plot)

ggsave(plot = cluster_plot, filename = "cluster_plot.svg", width = 202,
↪  height = 150, units = "mm", dpi = 300)
```